Welcome to EECS. 492 – Intro to AI!

Download Today's Slides & Set Up Poll Everywhere (PV)

Slides available on Canvas

If Poll Everywhere attendance doesn't work, come talk to me after class (so I can mark down your attendance)

Today's Class

- Meet the instructors
- What is AI?
- Course policies, syllabus, etc.
- A little bit of history

Meet the Staff

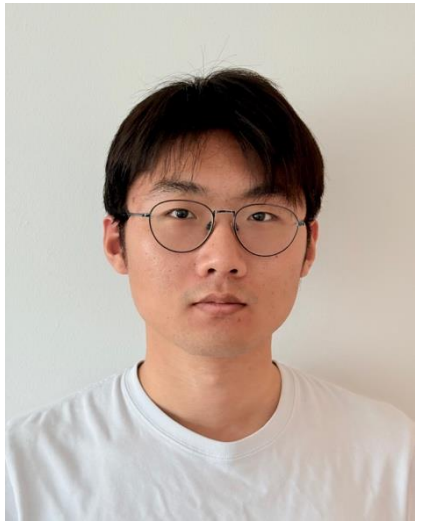
Alexander Rodríguez

- Assistant Professor in CSE
- Research focus
  - Time series & spatiotemporal modeling
  - AI for science
  - AI for high stake decision making
- Applications: Health, engineering, complex systems

alrodri.engin.umich.edu @_arodriguezca alrodri@umich.edu

Hometown: Lima, Peru

Course Staff



Ruipu Li (GSI)

Isabelle Baker (IA)

What is AI?
Multiple Views on Intelligence

When poll is active respond at **PollEv.com/492**

Turn to a person near you, introduce yourself, and spend a few minutes discussing the following question:

What is AI?

What is Artificial Intelligence?

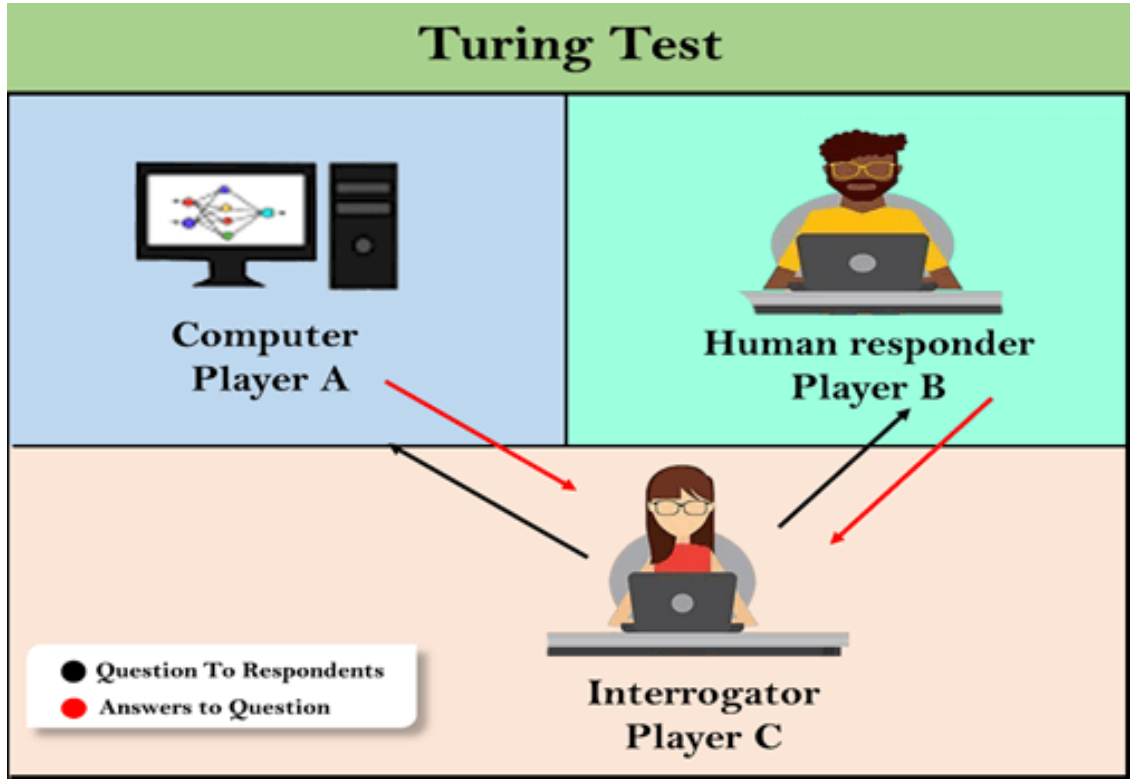
Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Acting Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- How can we define intelligence?
  - Turing Test

- Skills:
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine learning



https://www.javatpoint.com/turing-test-in-ai

Acting Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- Total Turing test = includes interacting with objects & people
  - Computer vision
  - Speech recognition
  - Robotics

- What are some issues with this test?

Acting Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

The quest for **"artificial flight"** succeeded when the engineers and inventors **stopped imitating birds** and started using wind tunnels and learning about aerodynamics. Aeronautical engineering texts do not define the goal of their field as making "machines that fly so exactly like pigeons that they can fool even other pigeons."

- Russell & Norvig, pg. 2 (our textbook)

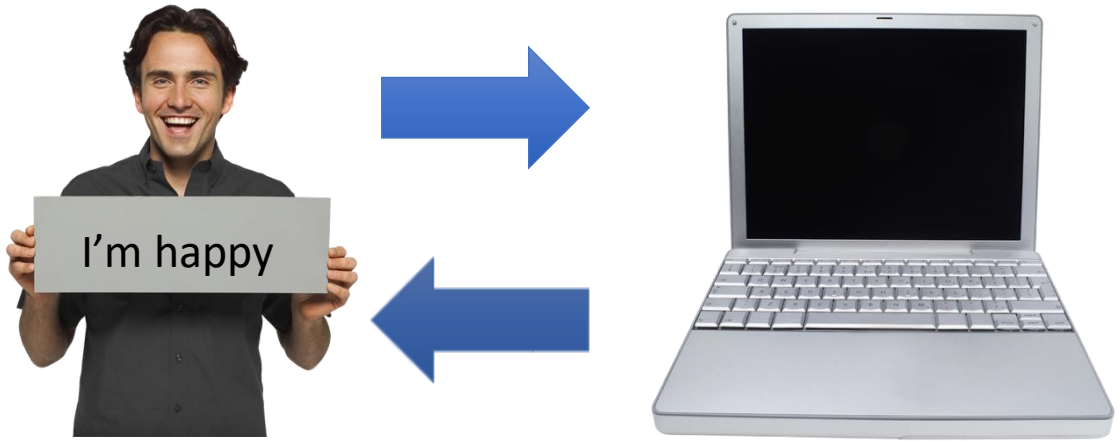
What is Artificial Intelligence?

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Thinking Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- First, we must know how humans think
  - Introspection
  - Psychological experiments
  - Brain imaging
- How can we assess if it does this?
- What can we learn about humans?
- **Cognitive science** – interdisciplinary field brings together computer AI models & experimental techniques from psychology



What is Artificial Intelligence?

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Thinking Rationally

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- Logic – provides precise notation for statements
- Example:
  - *Socrates is a man*
  - *All men are mortal*
  - Then, *Socrates is mortal*

- What are some limitations?

What is Artificial Intelligence?

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Acting Rationally

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- Agent – something that acts
  - What is a rational agent?
  - What is the goal of a rational agent?

- Rational agent approach – make the best decision given uncertainty

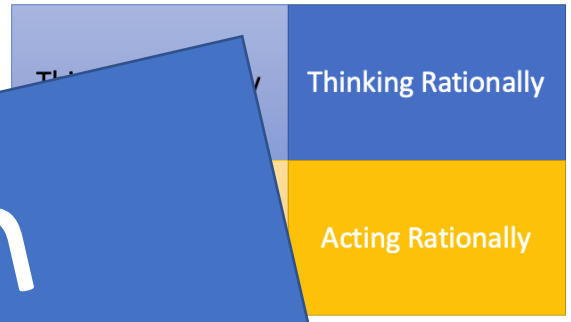
Acting Rationally

- Agent – somet

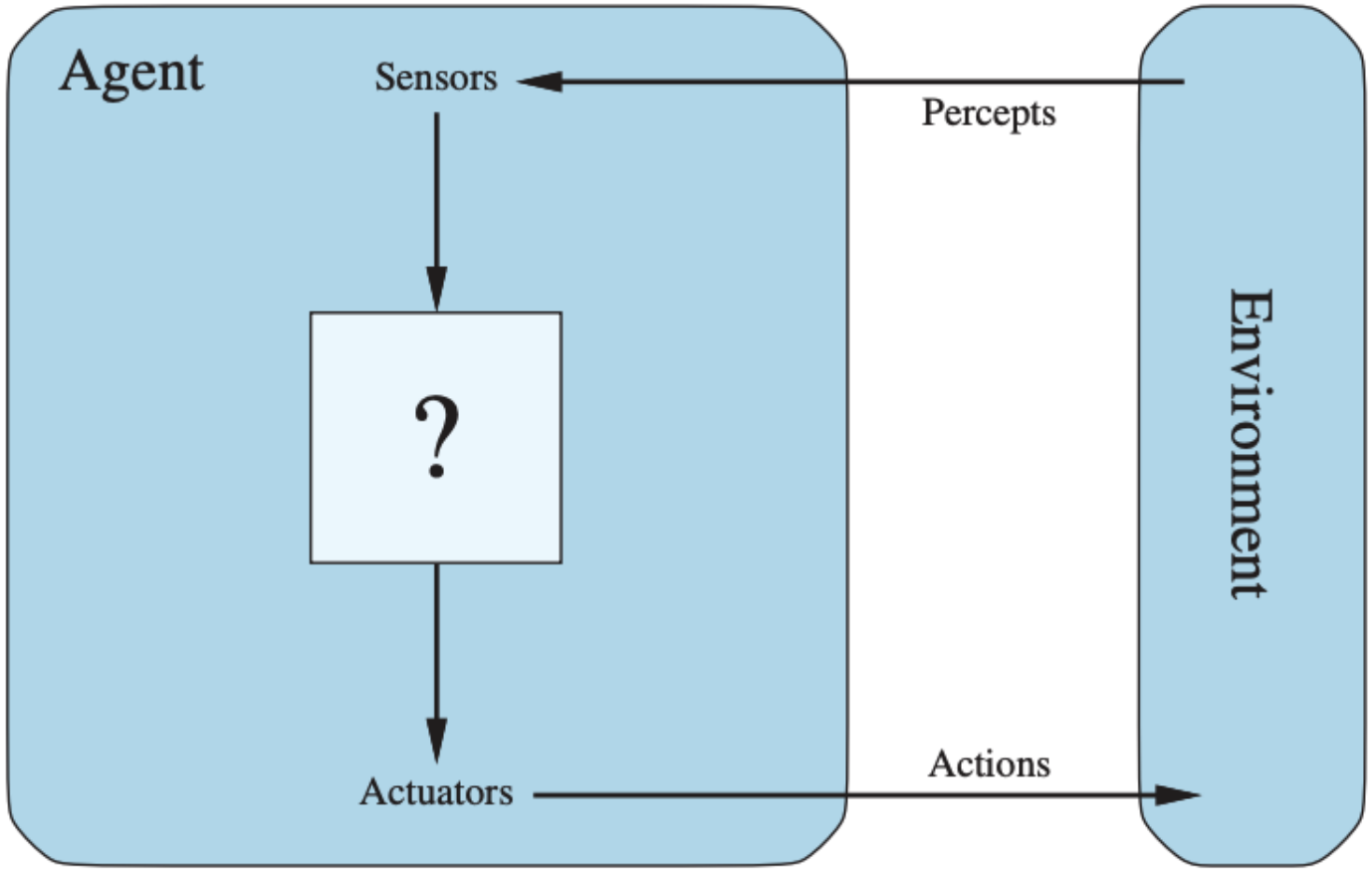
- R

uncertainty

Prevailing Approach throughout Much of AI's History



This Course: Designing Rational Agents



Beneficial Machines

- What's wrong with this prevailing approach?
  - Assumes that we can specify our objective completely
  - What is the objective for a self-driving car?

- Value alignment problem = values or objectives put into the machine must be aligned with those of the human.

- Solution: The machine is uncertain about what our objectives are
  - Machine should have incentives to act cautiously, ask permission, defer to human control

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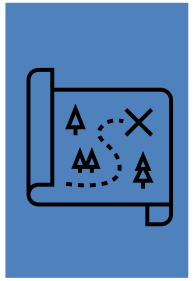
What questions about AI are you hoping to have answered this semester?

Break time

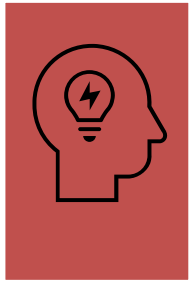
03:00

Course Policies, Syllabus, etc.

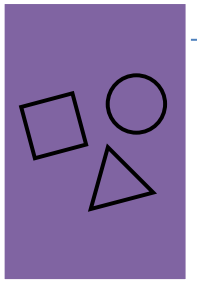
Big Themes of this Course



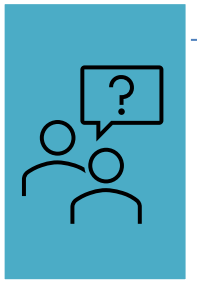
Search



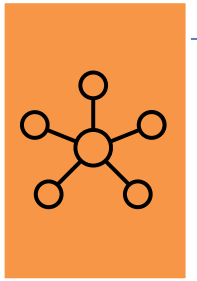
Problem Solving



Uncertainty



Learning



Logic

What is EECS 492?

- A first course in Artificial Intelligence,
  providing experience in a range of key ideas.

- Objectives – finish the course with:
  - Specific modeling and analytical skills (e.g., search, logic, probability, …)
  - Knowledge of many of the most important
    knowledge representation, reasoning, and machine learning schemes
  - General understanding of AI principles and practice

- Prepare for:
  - Further work in AI
  - Any work involving the design of computer programs for substantial applications

EECS 492 Topics

Boundaries are fuzzy!

A.I.

- Search
  - Uninformed
  - Informed
  - Local
  - Adversarial
- Constraint Satisfaction
- Logic
  - Propositional
  - First-Order
- Game theory

Machine Learning

- Decision Trees
- Linear Regression
- k-Nearest Neighbor
- SVMs
- Markov Decision Processes
- MCTS
- Reinforcement Learning

Neural Networks

- Feed forward (& deep)
- Convolutional
- Recurrent
- Transformers
- Generative

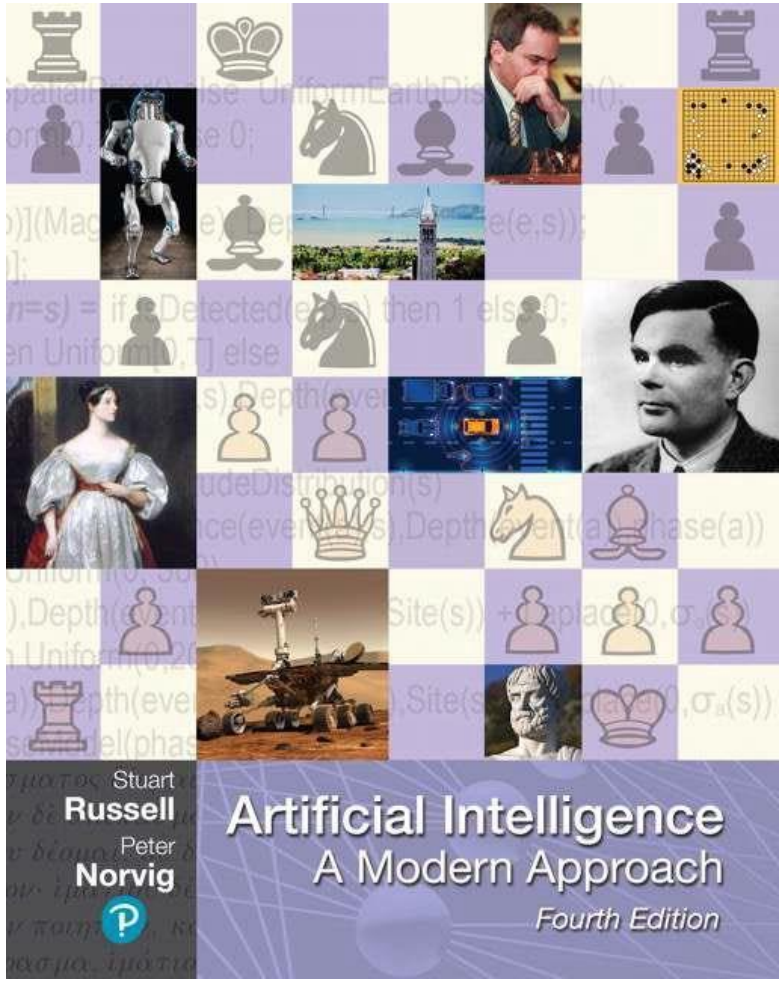
Topics subject to modest change

Lots of math across the entire course!

The Textbook

- Russell and Norvig:
  Artificial Intelligence: A Modern Approach
  (fourth edition)

- Encouraged, not required



Lectures

- Attendance will be taken using Poll Everywhere
  - Required in-person attendance: 14 out of the 27 lectures

- Built in flexibility – no exceptions will be granted except in extenuating circumstances

- Lectures will be recorded

Discussion Sections

- We want discussion sections to be a valuable part of this course for you!
  - Current events in AI; real-world problems and solutions
  - New exercises to practice the material
  - Review of old material (e.g., basic search algorithms, probability) that we will use in 492
  - Discussion section content won’t be a repeat of lecture content

- Ahead of time, we will tell you what discussion section will cover, so you know if it will be valuable for you to attend

Discussion Sections

- Attendance not required
- Discussion sections will **not** be recorded; slides will be available
- Try to join a consistent discussion section to get to know classmates and your instructor

Friday 11:30am-12:30pm	1024 FXB	Ruipu Li
Friday 12:30pm-1:30pm	1024 FXB	Ruipu Li
~~Friday 1:30pm-2:30pm~~	~~1024 FXB~~	

Office Hours

- Starting next week!

- Rodríguez's OH: Tuesday & Thursday 4:30-5:20 PM, BBB 3757
  - Come and talk about whatever's on your mind!
  - <u>Cannot help with programming</u>
  - E-mail to set up a different time / remote appointment

- Other staff office hours: to be listed in syllabus and course calendar

Staying Connected

- For **lecture content** questions that arise outside of class:
  ➜ Office hours
  and/or
  ➜ Piazza
- For questions about how to solve a **homework** problem:
  ➜ Office hours
  and/or
  ➜ Piazza if you can ask without sharing your answer and without sharing code
- For questions about course **policies**:
  ➜ Piazza if it's not personal
  or
  ➜ Admin form if it's personal.
- For concerns about how something was **graded**:
  ➜ See "Grading" section in the syllabus.
- For **personal** questions regarding:
  - Joining the course late
  - Sickness or another life challenge interfering with your work
  - SSD accommodations
  - Scheduling problems
  - Any other non-content-related personal question or concern
  - ➜ Admin form

CANVAS

piazza

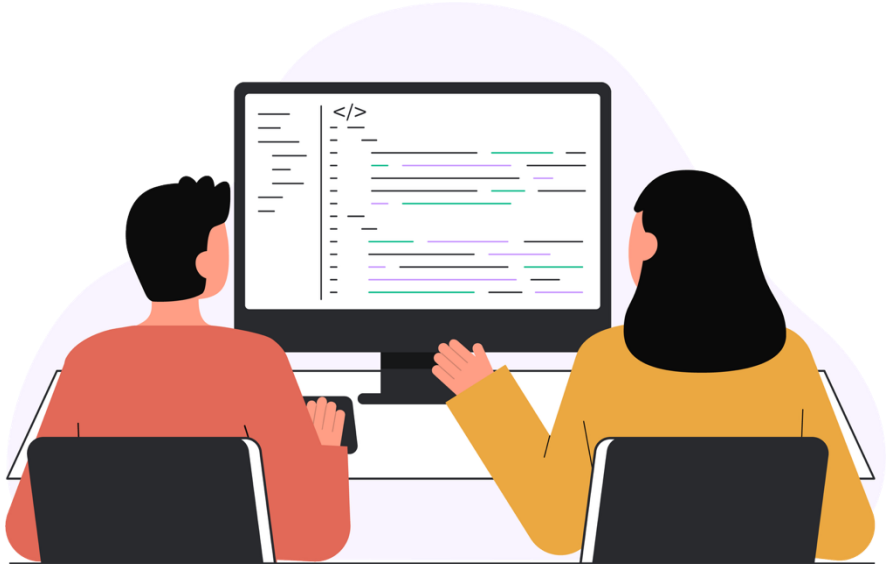
Staying Connected

- Canvas:
  - Slides
  - Homework assignments

- Piazza:
  - Main discussion forum, (much) faster than email!
  - Please post questions/ideas/thoughts there
  - You can post anonymously too (to your classmates)

Assignments

- Homeworks – Four throughout the semester
  - Conceptual questions and coding
  - Must **work alone** for <u>conceptual questions</u> (written part)
    - OK to discuss ideas and high-level concepts w/ classmates
  - Can **work with a partner** for <u>coding</u>



Assignments

- Ethics assignments – 2 throughout the semester
  - Required to read paper; you'll do a quiz before the discussion
  - Required to attend your discussion section
    - The section you enrolled to avoid overflow in the classroom
  - If you can't attend a discussion, must come to faculty office hours to talk about the reading within a weeek

Late Policy

- Late policy (for all assignments)
  - Minus 10% per day, up to 3 days

- Every student gets 5 late-day tokens to use for any HWs
  - No questions asked!
  - Hours are rounded up. So if you submit one HW 10hrs or 1hr late, it will be counted as 1 late day

- Medical or other emergencies are always excused separately: please fill out our admin form, always with documentation

Generative AI Policy

- Can use generative AI for conceptual questions
  - Need to include an explanation (see syllabus)

- CANNOT use generative AI for programming

Exams

- Both will be in person – make travel plans accordingly
- Midterm on Thursday, February 26, 6-8 pm (Tentative)
- Final on Thursday, April 23, 10:30 am-12:30 pm (Set by the university)

Wellness

- If for any reason you are having difficulty in this course, please come talk to me; I want to help.

Grading

- Homeworks 50% (12.5% each for 4 assignments)
- Ethics assignments 10% (5% each for 2 assignments)
- Midterm 15%
- Final 20%
- Attendance 5%
- Course evaluations +1% (extra credit, 0.5% for each evaluation – midterm and final)

Some advice on how to succeed in EECS 492

- Come to lectures regularly
  - Get to know the people around you so that you have friends to complete the homeworks with and study with
- Come to discussion sections regularly
  - Smaller setting to ask lots of questions
  - Get to know your GSI / IA
- If you're worried about falling behind, come to office hours regularly, even if you don't have specific questions
  - Use office hours as a time to work on your homework alongside other students

Some advice on how to succeed in EECS 492

- If something is causing you to fall behind in the class (a personal situation, an academic situation, etc.), come talk to me sooner rather than later!
  - I want you to succeed in this class, and I can help you make a plan to get back on track
- Start the assignments early – debugging takes time
- Start studying for the exams early

A Little Bit of History

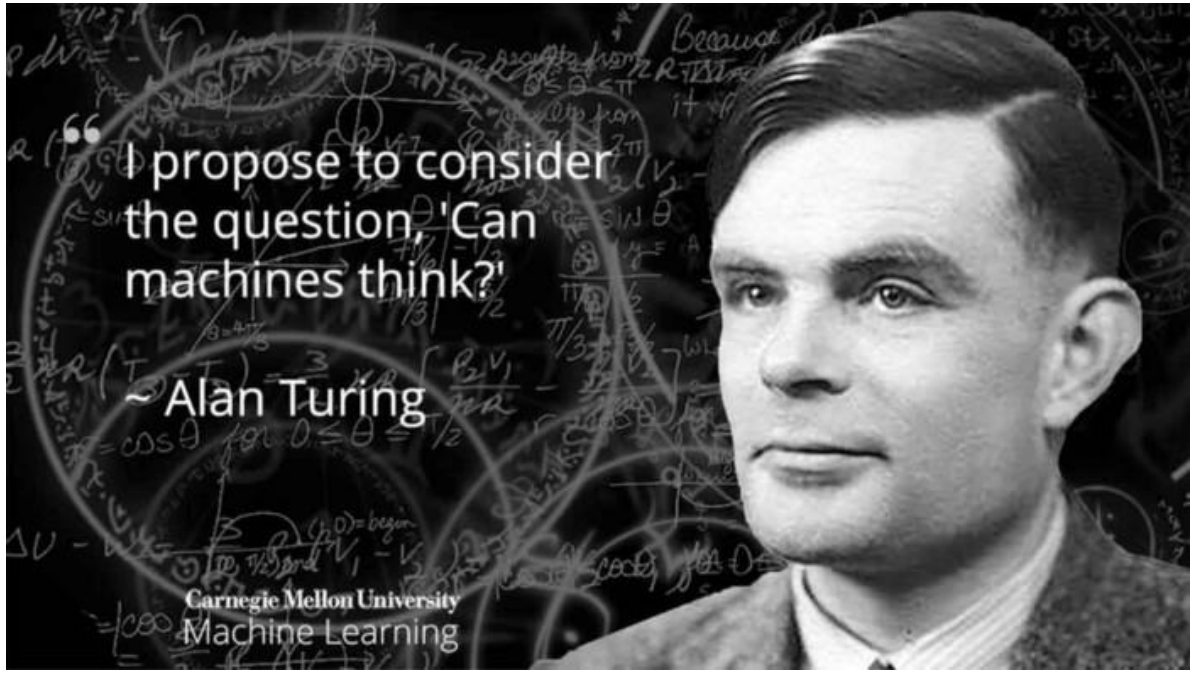
The beginning of AI…

- The neural network is born
  - 1943: Inspired by the brain, Warren McCulloch and Walter Pitts publish first mathematical model of a neural network
  - 1950: Marvin Minsky and Dean Edmonds build the first neural computer, the SNARC
    - 3,000 vacuum tubes
    - A surplus automatic pilot mechanism from a B-24 bomber
    - Network of only 40 neurons

http://cyberneticzoo.com/?m=200911&paged=3

The beginning of AI…

- Alan Turing's vision for AI
  - 1950 article – "Computing Machinery and Intelligence"
  - Introduced the Turing test, machine learning, genetic algorithms, reinforcement learning



The beginning of AI….

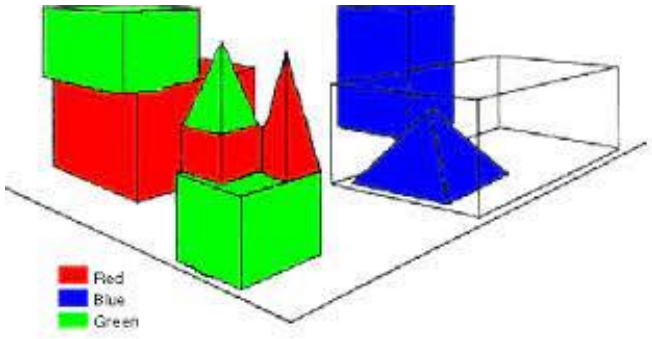
- Name “artificial intelligence” coined in 1956 Dartmouth conference:

  “… The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”

They didn’t quite finish…

“Look, Ma, no hands!”

- At the beginning, there was success was small, simple problems
  - 1952: Checkers-playing program performed at strong amateur level using early reinforcement learning (Arthur Samuel)
  - 1960s: Marvin Minsky worked on limited domains called **microworlds** (most famous: blocks world)



- 1958: John McCarthy creates Lisp programming language - dominant AI programming language for 30 years

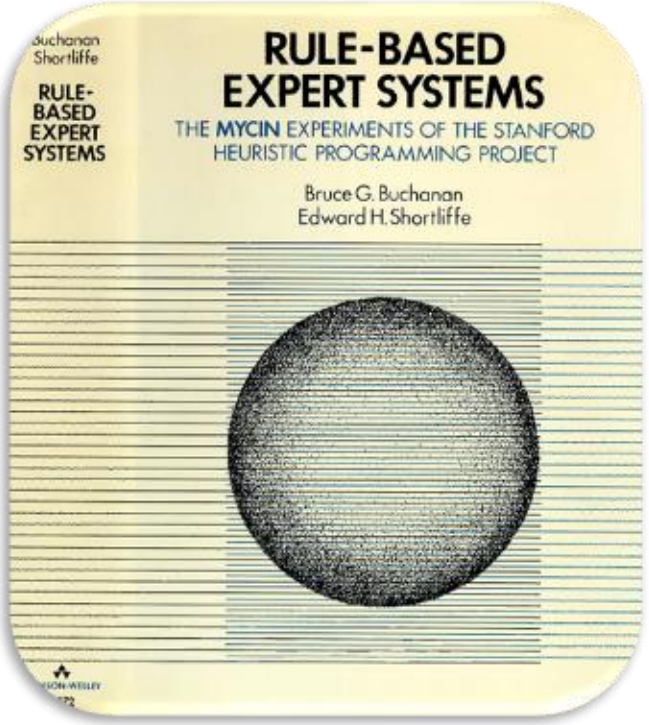
A Dose of Reality (1966-1973)

- Early programs had no subject knowledge
  - Success via simple syntactic manipulations
  - “The vodka is good but the meat is rotten.”

- Many problems were harder than they looked
  - Being able to solve something in principle (on a small example) doesn’t mean that you can solve it practically in the real world

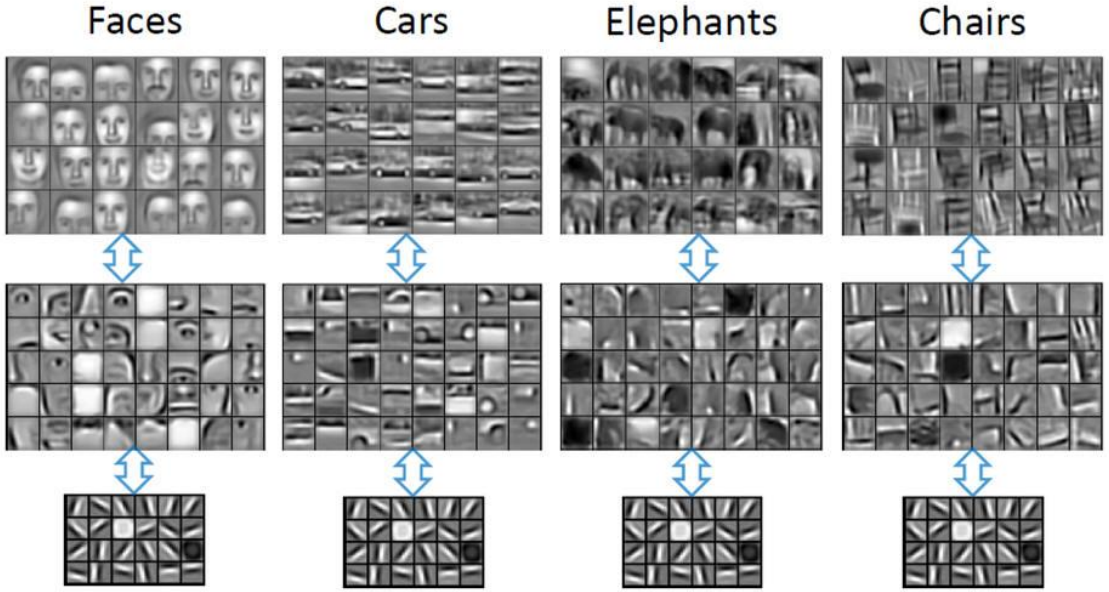
Expert Systems (1969-1986)

- Moving from general-purpose systems to **knowledge-intensive** systems in very specific domains
- Example: Mycin
  - 450 rules to diagnose blood infections
  - Issue – no general framework
  - Required expert knowledge from humans
  - Must capture uncertainty: introduced certainty factors
- Led to AI Winter: Companies failed to deliver on extravagant promises



The Return of Neural Networks (1986–now)

- Back-propagation was rediscovered in the mid-1980s!
  - Originally discovered in the early 1960s, but forgotten about
  - Enabled training of complex neural networks!



http://youqianhaozhe.com/research.htm

Probabilistic Reasoning and Machine Learning (1987-now)

- AI systems began to be grounded in more mathematical theory
  - Probability rather than Boolean logic
  - Machine learning rather than hand-coding
  - Experimental results rather than philosophical claims

- Large benchmark problem sets established

- AI began to draw more from other academic fields (e.g., statistics, decision theory)

Big Data (2001-now)

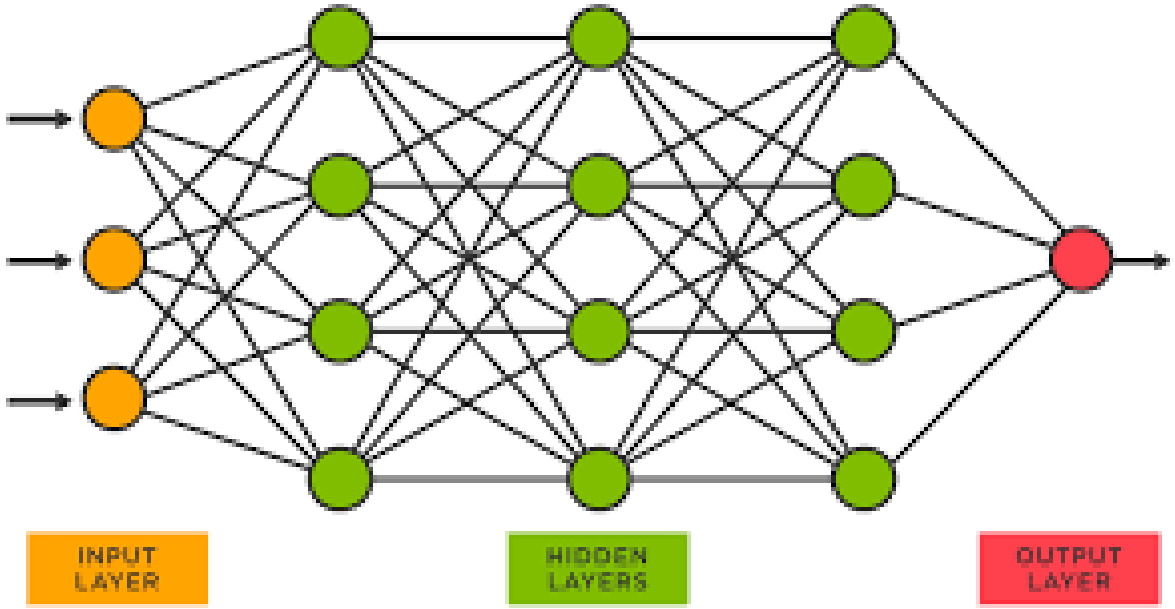
- Huge data sets are now available, thanks to the Internet and advances in computing power
  - Trillions of words of text
  - Billions of images
  - Billions of hours of speech video
  - So Much Data: genomic data, vehicle tracking data, clickstream data, …

- Banko and Brillo (2001) – improvement in performance from increasing the amount of data is 2 or 3 orders of magnitude more than improving the algorithm (!!!)

Deep Learning (2011-now)

- **Deep learning** – machine learning using multiple layers of simple, adjustable computing elements (neurons)

- Deep learning needs:
  - Powerful hardware
  - Large amounts of training data



https://www.tibco.com/

Where is AI Now?

- Human performance or better in chess, Go, poker, Pac-Man, Jeopardy!, skin cancer detection, prostrate cancer detection

Where is AI Now?

- Self-driving cars: Waymo vehicles have driven 10 million miles on public roads without a serious accident (human stepping in every 6,000 miles)

Final Reminders

- Office hours start next week (Tuesday)
- Read the syllabus
- We are looking forward to having a wonderful semester learning with all of you ☺